Detecting and Recovering from Deception in Model-Free Nash-Equilibrium Seeking: A Dual-Demodulation Analysis

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Abstract

We justify a model-free deception detector for Nash-equilibrium (NE) seeking, built entirely from the demodulation signal C_v each player already computes. Three facts are proved: (i) *blindness* — C_v vanishes at the deceptive equilibrium exactly as at an honest NE, so it cannot certify honesty; (ii) *detection* — a second demodulation C_v' at a probe frequency the deceiver does not inject into makes $D_v = |C_v' - C_v|$ equal to the injection magnitude, giving a threshold that separates the two regimes; (iii) *recovery* — re-randomizing the victim's probe frequency collapses the deceptive equilibrium and restores exponential convergence to the true NE. We delimit both statements. The detector is exactly evadable by a deceiver that mirrors its injection onto the second probe, at a cost of only $O(a^2)$ to the attack, so the guarantee is one of *probe secrecy* rather than of unconditional soundness; and recovery drives the deceiver's integral state out of any bounded set, so the post-switch dynamics terminate the deception in one of two ways — saturation or unbounded, trivially visible injection — rather than leaving δ_d irrelevant. We further show that the detector's residual is *not* monotone in the probing amplitude a: it decomposes into a curvature bias $O(a^2)$ and a ripple bias $O\big((k/a\omega)^2\big)$ that *grows* as a shrinks, so the correct design variable is k/ω and there is an optimal amplitude $a^\star \propto \sqrt{k/\omega}$. Numerical experiments confirm every identity and the residual scaling.

1 Setup

Two players, a victim v and a deceiver d, have scalar actions $x_v, x_d \in \mathbb{R}$ and costs $J_v(x_v, x_d)$, $J_d(x_v, x_d)$ (vector / N-player case: Remark 10). Actions carry sinusoidal probes,

$$x_v = u_v + a\sin(\omega_v t)\ \big(+\ a\sin(\omega_v' t)\big), \tag{1}$$

$$x_d = u_d + a\sin(\omega_d t) + a\,\delta_d(t)\sin(\omega_v t), \tag{2}$$

with probing amplitude $a > 0$; the optional ω_v'-probe is used only by the detector of §4. Each player passes its measured payoff through a washout (high-pass) filter with corner $\omega_h > 0$ before demodulating,

$$\dot z_i = -\omega_h\big(z_i - J_i(x)\big), \qquad h_i := J_i(x) - z_i, \qquad i \in \{v, d\}, \tag{3}$$

and updates its slow state by demodulating the filtered payoff at its own frequency,

$$\dot u_v = -\tfrac{2k}{a}\,h_v\sin(\omega_v t), \qquad \dot u_d = -\tfrac{2k}{a}\,h_d\sin(\omega_d t), \qquad k > 0. \tag{4}$$

Note that the victim's *update* uses only ω_v; the second probe enters the action (1) but drives no update, a point that matters in §5. The deceiver injects the victim's probe with amplitude driven by integral feedback toward a target payoff J_d^{ref}:

$$\dot{\delta}_d = \varepsilon\big(J_d(x) - J_d^{\mathrm{ref}}\big), \qquad \varepsilon > 0. \tag{5}$$

Setting $\delta_d \equiv 0$ recovers honest NE seeking. Partials $\partial_v J_v := \partial J_v/\partial x_v$, $\partial_d J_v := \partial J_v/\partial x_d$, etc. are evaluated at the slow state $u = (u_v, u_d)$, and $\overline{f} := \lim_{T\to\infty} \frac{1}{T}\int_0^T f\,dt$ is taken with u, δ_d frozen.

Remark 1 (Why the washout is not optional). Without (3) the update (4) is driven by J_v itself, whose DC component $J_v(u)$ contributes a term $-\frac{2k}{a}J_v(u)\sin(\omega_v t)$; although this averages to zero, it integrates to a ripple in u_v of amplitude $2k|J_v(u)|/(a\omega_v)$. Unless $\omega_v \gg 2k|J_v|/a^2$ this ripple exceeds the probe amplitude a and the loop destabilises — an a^{-2} requirement that is impractical for small a. The washout removes $J_v(u)$ and leaves a signal of size $O(a)$, reducing the ripple to $O(k/\omega)$; see Remark 6. Filtered extremum seeking in this form is standard [2].

Assumption 1 (Regularity)**.** $J_v, J_d \in C^3$ on an open operating set $U \subset \mathbb{R}^2$ containing the trajectories of interest.

Assumption 2 (Probe and filter selection)**.** The frequencies $\omega_v, \omega_v', \omega_d > 0$ are distinct and non-resonant: $m_1\omega_v + m_2\omega_v' + m_3\omega_d \neq 0$ for all integers with $0 < |m_1| + |m_2| + |m_3| \leq 3$. The washout corner is placed between the slow and fast time scales, $k \ll \omega_h \ll \omega_{\min} := \min\{\omega_v, \omega_v', \omega_d\}$.

Assumption 2 is the usual orthogonality condition of extremum seeking; it yields

$$\overline{\sin(\omega_i t)\sin(\omega_j t)} = \tfrac{1}{2}\mathbb{1}[\omega_i = \omega_j], \qquad \overline{\sin(\omega_i t)\sin(\omega_j t)\sin(\omega_\ell t)} = 0. \tag{6}$$

The next assumption fixes the adversary. It is stated here, once, because it delimits *both* the detector of §4 and the recovery of §6; §5 shows that neither survives its removal.

Assumption 3 (Probe secrecy)**.** The deceiver knows ω_v and injects at that frequency only. It neither knows nor injects at any other frequency the victim uses — in particular not at the monitoring probe ω_v' of §4, nor at the post-switch frequency ω_v'' of §6.

Assumption 4 (Bounded injection)**.** The injection amplitude is bounded: there is $\bar{\delta} < \infty$ with $|\delta_d(t)| \leq \bar{\delta}$ for all t, and the corresponding trajectories remain in U. Equivalently, (5) carries a saturation or anti-windup element at level $\bar{\delta}$.

Assumption 4 is vacuous while deception succeeds — at a DNE the integrator (5) is at rest, so δ_d is automatically bounded — and it is *not* vacuous after recovery, which is precisely the content of Corollary 7. Constants written $O(a^2)$ below depend on $\bar{\delta}$ through the second- and third-order Taylor coefficients of J_v evaluated on the enlarged perturbation $a(1+\bar{\delta})$; this dependence is uniform on U by Assumption 1.

Averaging and the order of limits. We analyse the averaged system obtained below and record the probing bias as $O(a^2)$. Transfer to the true time-varying system is by standard averaging [3, Ch. 10], but the quantifiers matter and are stated here once, since they are the source of a design rule that is easy to get backwards: *for each fixed $a > 0$ there exists $\bar{\omega}(a) < \infty$ such that for probe frequencies exceeding $\bar{\omega}(a)$ the true trajectories remain $O(a)$-close to the averaged ones*, uniformly in t under the stability hypothesis of Theorem 6. The threshold $\bar{\omega}$ *depends on* a and grows as $a \downarrow 0$. Consequently one may not fix ω and then send $a \to 0$; doing so eventually violates the averaging hypothesis, and the resulting error is quantified in Remark 6.

2 Averaged dynamics and the deceptive equilibrium

Everything rests on the demodulation of the victim's filtered payoff at a frequency ω. Because the washout (3) has transfer function $s/(s+\omega_h)$, a component of J_v at frequency ω is attenuated by $\omega/\sqrt{\omega^2+\omega_h^2}$ and phase-shifted by $\arctan(\omega_h/\omega)$; demodulating with $\sin(\omega t)$ therefore retains the fraction

$$\Gamma(\omega) := \frac{\omega^2}{\omega^2+\omega_h^2} \in (0,1). \tag{7}$$

We define the *compensated* demodulation

$$\mathcal{D}_\omega[J_v] := \frac{2}{a\,\Gamma(\omega)}\,\overline{h_v \sin(\omega t)}. \tag{8}$$

Compensation uses only ω_h and the victim's own probe frequency, both of which it chooses, so (8) remains model-free. Up to the factor $\Gamma(\omega_v)$ this is exactly the quantity the victim already forms in (4): on average $\dot{u}_v = -k\,\Gamma(\omega_v)\,\mathcal{D}_{\omega_v}[J_v]$.

Lemma 1 (Demodulation returns a perceived gradient)**.** *Under Assumptions 1–2 and 4, with the deceiver injecting $a\delta_d \sin(\omega_v t)$:*

(a) $C_v := \mathcal{D}_{\omega_v}[J_v] = \partial_v J_v(u) + \delta_d\,\partial_d J_v(u) + O(a^2)$;

(b) if the victim also probes with amplitude a at $\omega_v' \notin \{\omega_v, \omega_d\}$ and Assumption 3 holds, then $C_v' := \mathcal{D}_{\omega_v'}[J_v] = \partial_v J_v(u) + O(a^2)$.

Proof. With u, δ_d frozen, a third-order Taylor expansion of J_v about u gives

$$J_v(x) = J_v(u) + \partial_v J_v\,(x_v - u_v) + \partial_d J_v\,(x_d - u_d) + R_2, \qquad R_2 = O(a^2),$$

where $x_v - u_v$ and $x_d - u_d$ are read off (1)–(2) and the implied constants are uniform in δ_d by Assumption 4. The washout is linear and annihilates the constant $J_v(u)$; each remaining sinusoidal component at frequency ω survives with gain $\Gamma(\omega)^{1/2}$ and phase $\arctan(\omega_h/\omega)$, so that after demodulation at that same frequency the net factor is $\Gamma(\omega)$, which (8) divides out.

(a) Multiply h_v by $\frac{2}{a\Gamma(\omega_v)}\sin(\omega_v t)$ and average. By (6),

$$\tfrac{2}{a}\partial_v J_v \overline{(x_v - u_v)\sin(\omega_v t)} = \tfrac{2}{a}\partial_v J_v \cdot a\tfrac{1}{2} = \partial_v J_v,$$

$$\tfrac{2}{a}\partial_d J_v \overline{(x_d - u_d)\sin(\omega_v t)} = \tfrac{2}{a}\partial_d J_v \big(a\underbrace{\overline{\sin\omega_d t\,\sin\omega_v t}}_{=0} + a\delta_d \underbrace{\overline{\sin^2\omega_v t}}_{=1/2}\big) = \delta_d\,\partial_d J_v.$$

Each second-order term of R_2 is a product of two probe sinusoids times $\sin(\omega_v t)$, hence averages to zero by (6) regardless of its individual filter gain; the third-order remainder is $O(a^3)$ and contributes $\frac{2}{a}O(a^3) = O(a^2)$.

(b) Multiply instead by $\frac{2}{a\Gamma(\omega_v')}\sin(\omega_v' t)$. The $\partial_v J_v$ term again yields $\partial_v J_v$ via the victim's own ω_v'-probe, while by Assumption 3 the deceiver injects nothing at ω_v', so $\overline{(x_d - u_d)\sin(\omega_v' t)} = 0$ because $\omega_v' \neq \omega_d, \omega_v$. The $\partial_d J_v$ contribution therefore drops. □

Remark 2 (Compensation is necessary, not cosmetic). If the raw demodulations $\frac{2}{a}\overline{h_v \sin(\omega t)}$ are used instead of (8), Lemma 1 returns $\Gamma(\omega_v)\big(\partial_v J_v + \delta_d \partial_d J_v\big)$ and $\Gamma(\omega_v')\partial_v J_v$. Since $\Gamma(\omega_v) \neq \Gamma(\omega_v')$, the true gradient no longer cancels in the difference and the detector of §4 acquires the spurious floor

$$\big|\Gamma(\omega_v') - \Gamma(\omega_v)\big|\,\big|\partial_v J_v\big| = \omega_h^2 \Big|\tfrac{1}{\omega_v^2} - \tfrac{1}{\omega_v'^2}\Big|\,\big|\partial_v J_v\big| + O(\omega_h^4),$$

which does not vanish with a and is largest exactly where the detector must be trusted.

Lemma 1(a) is the mechanism of deception: $\dot{u}_v = -k\Gamma(\omega_v)C_v$ drives u_v to a zero of the *perceived* gradient $\partial_v J_v + \delta_d \partial_d J_v$, not of $\partial_v J_v$. Absorbing $\Gamma(\omega_v) = 1 + O(\omega_h^2/\omega_v^2)$ into k, the averaged closed loop is

$$\dot{u}_v = -k\big(\partial_v J_v(u) + \delta_d \partial_d J_v(u)\big) + O(a^2), \quad \dot{u}_d = -k\,\partial_d J_d(u) + O(a^2), \quad \dot{\delta}_d = \varepsilon\big(J_d(u) - J_d^{\mathrm{ref}}\big) + O(a^2). \tag{9}$$

Definition 1 (Honest and deceptive NE)**.** $u^\star$ is a true NE if $G_{\mathrm{hon}}(u^\star) = 0$, where $G_{\mathrm{hon}}(u) := \big(\partial_v J_v(u),\ \partial_d J_d(u)\big)$. A triple (u^*, δ_d^*) with $\delta_d^* \neq 0$ is a *deceptive Nash equilibrium* (DNE) if it is an equilibrium of (9):

$$\partial_v J_v(u^*) + \delta_d^* \partial_d J_v(u^*) = 0, \qquad \partial_d J_d(u^*) = 0, \qquad J_d(u^*) = J_d^{\mathrm{ref}}. \tag{10}$$

The first identity is the crux: $\partial_v J_v(u^*) = -\delta_d^* \partial_d J_v(u^*)$, so the victim's true gradient at the DNE is generically nonzero, of magnitude $|\delta_d^*||\partial_d J_v(u^*)|$. The third identity is what makes deception worth running — J_d^{ref} is chosen strictly better for the deceiver than the honest value $J_d(u^\star)$ — and it is what makes the integrator diverge once the victim escapes (Corollary 7).

3 Blindness of the raw signal

Lemma 2 (Single-frequency blindness)**.** *Along* (9), $C_v = -\frac{1}{k}\dot{u}_v + O(a^2)$. *Hence* $C_v = O(a^2)$ *at every equilibrium of the victim's slow dynamics — at an honest NE* ($\delta_d = 0$, $\partial_v J_v(u^\star) = 0$) *and at a DNE* ($\delta_d = \delta_d^* \neq 0$, *perceived gradient zero by* (10)) *alike. No test on* C_v *alone can separate the two regimes.*

Proof. The identity is (9). In both cases the bracketed term vanishes, so $\dot{u}_v = O(a^2)$ and the two limits coincide. □

The same argument kills the action-based residual $r_v = x_v(t{+}1) - \hat{x}_v(t{+}1)$: the victim *does* follow the honest update rule, only its input J_v is corrupted, so $r_v = O(a^2)$ by construction. The defect is not the residual idea but the use of a single compromised frequency.

4 A detector from a second demodulation

Take a second copy of (8) at the monitoring frequency ω_v' and set

$$D_v := \big|C_v' - C_v\big|. \tag{11}$$

Theorem 3 (Detection identity)**.** *Under Assumptions 1–4, along* (9),

$$D_v = |\delta_d|\,\big|\partial_d J_v(u)\big| + O(a^2) \qquad \textit{at every time } t. \tag{12}$$

Proof. Subtracting Lemma 1(a) from (b), $C_v' - C_v = -\delta_d \partial_d J_v(u) + O(a^2)$: the true gradient $\partial_v J_v(u)$ appears in both demodulations and cancels, leaving exactly the injection term that C_v could not isolate. Take absolute values. □

Remark 3 (Why the difference and not C_v' alone). At the DNE $C_v = O(a^2)$ by Lemma 2, so D_v and $|C_v'|$ agree *there*. They do not agree during honest transients, where $C_v' = C_v \neq 0$ and hence $D_v = O(a^2)$ while $|C_v'|$ is as large as the honest gradient. Testing $|C_v'|$ therefore raises an alarm on every honest run that has not yet converged; the subtraction is what buys a detector valid at all times rather than only after settling, which is the content of the phrase "at every time t" in (12).

Assumption 5 (Effective deception)**.** There is $c_0 > 0$ with $|\partial_d J_v(u)| \geq c_0$ for all $u \in U$. Deception is *active* at time t if $|\delta_d(t)| \geq \delta_{\min}$ for a fixed $\delta_{\min} > 0$.

This only says deception is effective: if it were not, there would be nothing to detect.

Corollary 4 (Soundness, completeness, separation)**.** *Let Assumptions 1–5 hold — in particular probe secrecy, Assumption 3 — and let*

$$\eta(a, k, \omega) := \sup_t \Big| D_v - |\delta_d| |\partial_d J_v(u)| \Big| \tag{13}$$

denote the total residual of (12) on the true system. Suppose the design parameters are chosen so that

$$\eta(a, k, \omega) \leq \tfrac{1}{4}\, \delta_{\min} c_0, \tag{14}$$

and fix $\rho := \frac{1}{2}\delta_{\min} c_0$. Then under honest play $D_v \leq \frac{1}{4}\delta_{\min} c_0 < \rho$ for all t; whenever deception is active $D_v \geq \delta_{\min} c_0 - \frac{1}{4}\delta_{\min} c_0 = \frac{3}{4}\delta_{\min} c_0 > \rho$. Hence the rule "$D_v > \rho \Rightarrow$ deception, $D_v \leq \rho \Rightarrow$ honest" has neither false alarms nor missed detections against a deceiver satisfying Assumption 3.

Proof. Honest play sets $\delta_d \equiv 0$ in (12), leaving $D_v \leq \eta$ within the bound (14). Active deception gives $|\delta_d||\partial_d J_v| \geq \delta_{\min} c_0$ by Assumption 5; subtracting η yields the lower bound. The ranges $[0, \rho)$ and (ρ, ∞) are disjoint. □

5 The detector's attack surface

Corollary 4 is a statement about a deceiver that injects on one frequency. That restriction is not cosmetic: dropping it defeats the detector exactly, at negligible cost to the attack.

Proposition 5 (Exact evasion by a mirroring deceiver)**.** *Suppose the deceiver injects on both of the victim's probe frequencies,*

$$x_d = u_d + a \sin(\omega_d t) + a\, \delta_d(t) \sin(\omega_v t) + a\, \delta'(t) \sin(\omega_v' t),$$

in violation of Assumption 3. Then

$$D_v = \big|\delta' - \delta_d\big|\, \big|\partial_d J_v(u)\big| + O(a^2). \tag{15}$$

In particular the choice $\delta' \equiv \delta_d$ gives $D_v = O(a^2)$ for all t: the detector reads honest while deception is active with arbitrary strength. Moreover this evasion costs the deceiver nothing at first order — the victim's update (4) demodulates at ω_v only, so the added component averages out of $\dot{u}_v$ and the DNE (10) is unchanged to $O(a^2)$; the deceiver's own payoff acquires only the $O(a^2)$ probing bias of the extra perturbation.

Proof. Repeat Lemma 1(b) with the extra component present. Demodulating at ω_v', the term $\frac{2}{a}\partial_d J_v \cdot a\delta' \overline{\sin^2 \omega_v' t} = \delta' \partial_d J_v$ now survives, so $C_v' = \partial_v J_v + \delta' \partial_d J_v + O(a^2)$, while C_v is unchanged. Subtracting gives (15). For the cost claim, demodulating the added component at ω_v gives $\overline{\sin(\omega_v' t)\sin(\omega_v t)} = 0$ by (6), so it does not enter the averaged victim dynamics (9); it enters J_d only through second-order Taylor terms, contributing $O(a^2)$ to (5). □

Proposition 5 locates the guarantee precisely. Detection here is not an information-theoretic impossibility result for the deceiver; it is a *secrecy* statement, and its strength is exactly the strength of the victim's ability to keep ω_v' unknown. Two consequences are worth recording.

Remark 4 (Secrecy is a timing condition, not a static one). The deceiver's payoff J_d depends on x_v through $\partial_v J_d \neq 0$ — which is precisely what makes deception profitable in the first place — so a deceiver that spectrally scans its own payoff can in principle recover ω_v' after an acquisition time T_{acq} determined by its measurement noise and the probe amplitude a. Assumption 3 is therefore realistic only if the victim re-draws ω_v' on a timescale T_{hop} with

$$\frac{1}{\omega_{\min}} \ll T \ll T_{\text{hop}} < T_{\text{acq}},$$

where T is the detection window of Remark 7. The middle inequality says the detector must complete a decision between re-draws; the right one says the deceiver must not finish acquiring first. Both sides are quantifiable and neither is addressed here.

Remark 5 (Toward a randomized detector). A natural strengthening is to give the victim a bank of m monitoring probes and to draw the one driving (4) at random. Blinding the resulting $\binom{m}{2}$ pairwise discrepancies requires the deceiver to mirror its injection across the whole bank, which multiplies its action perturbation by m and so is no longer free once the action is amplitude-constrained; mirroring only a subset leaves at least one pair with $\delta' \neq \delta_d$ and hence, by (15), a nonzero D_v. This converts the deterministic guarantee of Corollary 4 into a detection probability with an explicit security parameter m, at the cost of m additional non-resonant frequencies — a scarce resource in N-player games, where Assumption 2 must hold across all players simultaneously. We leave the analysis to future work.

Remark 6 (The residual is not monotone in a; design rule). It is tempting to read (12) as saying that (14) is met by taking a small. This is false on the true system, and the error is in the wrong direction. Two mechanisms contribute to η.

(i) *Curvature bias*, from the third-order Taylor remainder in Lemma 1: $O(a^2)$, decreasing in a.

(ii) *Ripple bias.* In (4) the prefactor $2k/a$ multiplies a filtered payoff of size $h_v = O(a)$, so $\dot{u}_v = O(k)$ *independently of* a; its oscillatory part, at the mixed frequencies $\omega_i \pm \omega_j$, integrates to a ripple in u_v of amplitude $O(k/\omega_{\min})$, again independent of a. What matters for the demodulation is the ripple *relative* to the probe,

$$\nu := \frac{\text{ripple}}{a} = O\Big(\frac{k}{a\,\omega_{\min}}\Big),$$

which *grows* as $a \downarrow 0$. Its leading contribution to η is $O(\nu^2)$.

Hence

$$\eta \lesssim C_1 a^2 + C_2\Big(\frac{k}{a\,\omega_{\min}}\Big)^2, \tag{16}$$

which is minimised not at $a \to 0$ but at

$$a^\star = \Big(\frac{C_2}{C_1}\Big)^{1/4}\sqrt{\frac{k}{\omega_{\min}}}, \qquad \eta_{\min} = 2\sqrt{C_1 C_2}\,\frac{k}{\omega_{\min}}. \tag{17}$$

The design condition (14) is therefore a condition on $k/\omega_{\min}$, not on a alone:

$$\frac{k}{\omega_{\min}} \lesssim \frac{\delta_{\min} c_0}{8\sqrt{C_1 C_2}}.$$

This is consistent with, and is the quantitative form of, the order-of-limits caveat in §1: shrinking a at fixed ω eventually violates the averaging hypothesis. Section 7 reports $\eta \approx 0.19\,\nu^2$ for the test game, with $C_1 a^2$ negligible over the range examined.

Remark 7 (Finite averaging window). An implementation replaces $\bar{\cdot}$ by a sliding window $\frac{1}{T}\int_{t-T}^{t}$, which leaves a residual ripple $O(1/(\omega_{\min}T))$ in D_v in addition to (16). The window must therefore satisfy $T \gg 1/\omega_{\min}$ while remaining short enough that u is approximately frozen over it, $T \ll 1/k$; the interval is nonempty precisely under the time-scale separation of Assumption 2. Identity (12) additionally requires δ_d to be slow relative to the window, $\varepsilon \ll 1/T$, or the two demodulations are compared at effectively different injection amplitudes.

Remark 8. D_v uses only the victim's own payoff samples and its own probes at ω_v, ω_v' — no model of J_v, no knowledge of the deceiver's identity, of δ_d, or of J_d. It is admissible in the same model-free setting as the NE-seeking dynamics themselves. The constant c_0 enters only the calibration of ρ, which may equivalently be set from the empirical level of D_v during a known-honest interval.

6 Recovery: collapsing the deceptive equilibrium

Detection and recovery are two uses of the same primitive. At a trigger time t_0 (when $D_v > \rho$) the victim re-randomizes the frequency driving (4) from ω_v to a secret $\omega_v'' \notin \{\omega_v, \omega_d\}$ satisfying Assumptions 2 and 3.

Assumption 6 (Stable true NE)**.** The honest pseudogradient flow $\dot{u} = -kG_{\text{hon}}(u)$ has an equilibrium $u^\star$ (Definition 1) at which $-k\,DG_{\text{hon}}(u^\star)$ is Hurwitz (e.g. strongly monotone games).

Theorem 6 (Recovery)**.** *Let Assumptions 1, 2, 3, 4, 6 hold. There exists $\bar{a} > 0$, depending on $\bar{\delta}$, such that* for every $a \in (0, \bar{a})$ *there exist $\bar{\omega}(a) < \infty$ and $\bar{\varepsilon}(a) > 0$ with the following property: if the probe frequencies exceed $\bar{\omega}(a)$ and $\varepsilon \in (0, \bar{\varepsilon}(a))$, then after the switch $\omega_v \mapsto \omega_v''$ at t_0 the slow state $u(t)$ converges exponentially to an $O(a)$-neighborhood of $u^\star$, for every trajectory of $\delta_d(t)$* admissible under Assumption 4*; in particular $x_v(t) \to u_v^\star + a\sin(\omega_v'' t) + O(a)$.*

Proof. The victim now demodulates and updates at ω_v'' while the deceiver, by Assumption 3, still injects at ω_v. Repeating Lemma 1(a) with ω_v'' in the update and the injection left at ω_v, the cross term carries $\overline{\sin(\omega_v t)\sin(\omega_v'' t)} = 0$ by (6). The perceived gradient thus loses its bias and $\dot{u} = -kG_{\text{hon}}(u) + O(a^2)$; crucially δ_d no longer appears in the u-dynamics, so the deceiver's integral state (5) is decoupled from u.

The decoupling alone does not make δ_d irrelevant, because the $O(a^2)$ term is uniform in δ_d only on bounded sets: the neglected Taylor remainders carry factors $a^2\delta_d^2\,\partial_{dd}^2 J_v$ and $a^3\delta_d^3\,\partial^3 J_v$, so an unbounded δ_d would destroy both the perturbation bound and the containment of x_d in U required by Assumption 1. Assumption 4 supplies the needed uniformity: on $|\delta_d| \le \bar{\delta}$ the remainder is bounded by $C(\bar{\delta})a^2$ with C independent of t.

By Assumption 6, $u^\star$ is locally exponentially stable for $\dot{u} = -kG_{\text{hon}}(u)$; with a quadratic Lyapunov function V satisfying $\dot{V} \le -c\|u - u^\star\|^2$ near $u^\star$, the $C(\bar{\delta})a^2$ probing bias is a vanishing perturbation, so for $a < \bar{a}(\bar{\delta})$ the averaged system converges exponentially to an $O(a^2)$-ball around $u^\star$. By the averaging theorem [3, Thm. 10.4], this exponential stability transfers to the true time-varying system up to an $O(a)$-neighborhood once the probe frequencies exceed a threshold $\bar{\omega}(a)$ determined by the chosen a; the small-ε condition places the decoupled integrator (5) in standard singular-perturbation form [3, Ch. 11] and does not affect the u-limit. □

Assumption 4 is not a technical convenience: without it the post-switch dynamics do not merely fail to satisfy the hypotheses of Theorem 6, they actively violate them, and in a way that is itself informative.

Corollary 7 (Post-recovery windup)**.** *Suppose the deceiver runs (5) without saturation and the conclusion of Theorem 6 holds on $[t_0, t_1]$, so that $J_d(u(t)) \to J_d(u^\star)$. Assume the deception was worth running, $\gamma := J_d(u^\star) - J_d^{\mathrm{ref}} \neq 0$. Then for every $\theta \in (0,1)$ there is t_1 with*

$$\big|\delta_d(t)\big| \;\geq\; \big|\delta_d(t_1)\big| + \theta\,\varepsilon\,|\gamma|\,(t - t_1), \qquad t \geq t_1,$$

so δ_d leaves every bounded set in finite time and the deceiver's action perturbation $a|\delta_d|$ grows without bound.

Proof. By Theorem 6, $u(t) \to u^\star$ up to $O(a)$, hence $J_d(u(t)) - J_d^{\mathrm{ref}} \to \gamma$ by continuity (Assumption 1); choose t_1 so that $|J_d(u(t)) - J_d^{\mathrm{ref}}| \geq \theta|\gamma|$ with fixed sign for $t \geq t_1$. Integrating (5) gives the bound. The conclusion of Theorem 6 is unaffected by the growth of δ_d over $[t_1, t]$ only as long as x_d remains in U; the corollary asserts that this eventually fails. □

Remark 9 (Recovery terminates the deception either way). Corollary 7 closes the loop that Theorem 6 leaves open. A deceiver whose integrator is unsaturated cannot simply persist after the victim hops: its injection amplitude diverges linearly at rate $\varepsilon|\gamma|$, and an injection of unbounded amplitude at a known frequency is detectable by any spectral monitor, quite apart from the actuation limits it violates. A deceiver that does saturate at $\bar{\delta}$ satisfies Assumption 4, and Theorem 6 applies verbatim: the victim converges to $u^\star$ while the injection persists uselessly at a frequency no one demodulates. A deceiver that abandons the attack sets $\delta_d \to 0$ and the game is honest. In all three cases the deception ends; the asymmetry the deceiver exploited — a single influence channel at a frequency the victim was obliged to keep using — is gone the moment the victim is free to move.

Remark 10 (Vector and N-player extension). For $x_v \in \mathbb{R}^{n_v}$ and several deceivers, replace $\partial_v J_v, \partial_d J_v$ by the Jacobian blocks $\nabla_v J_v, \nabla_d J_v$, the scalar δ_d by the injection-amplitude vector δ, and absolute values by Euclidean norms. Lemma 1 holds componentwise, (12) becomes $D_v = \|(\nabla_d J_v)\delta\| + O(a^2)$, Assumption 5 becomes the coercivity bound $\|(\nabla_d J_v)\delta\| \geq c_0\|\delta\|$ on U, Assumption 4 becomes $\|\delta\| \leq \bar{\delta}$, and Assumption 6 becomes Hurwitzness of $-k\,DG_{\mathrm{hon}}(u^\star)$. Theorem 3, Corollary 4, Theorem 6 and Corollary 7 carry over; Proposition 5 becomes $D_v = \|(\nabla_d J_v)(\delta' - \delta)\| + O(a^2)$, so evasion requires the mirroring condition $(\nabla_d J_v)(\delta' - \delta) = 0$, which for $n_v < n_d$ admits solutions with $\delta' \neq \delta$ — the deceiver need only match the injection modulo $\ker(\nabla_d J_v)$.

7 Numerical validation

We integrate the true time-varying system (1)–(5) (not the averaged one) on the strongly monotone quadratic game

$$J_v = \tfrac{1}{2}(x_v - 1)^2 + c\,x_v x_d + e\,x_d, \qquad J_d = \tfrac{1}{2}(x_d - 1)^2 + c\,x_v x_d, \qquad c = 0.3,\; e = 0.5,$$

with $a = 0.05$, $k = 0.3$, $\varepsilon = 0.08$, probe frequencies $2\pi\{5,\, 7.3,\, 11.7\}$, $\omega_h = 2\pi$, $J_d^{\mathrm{ref}} = 0.15$, sliding window $T = 10$.

quantity	predicted	measured
DNE action x_v^*	0.5445	0.5424
injection amplitude δ_d^*	0.308	0.302
$\lvert\partial_v J_v(u^*)\rvert$ vs. $\lvert\delta_d^*\rvert\lvert\partial_d J_v(u^*)\rvert$, cf. (10)	0.2057	0.2061
Lemma 1(a) residual $\lvert C_v - (\partial_v J_v + \delta_d \partial_d J_v)\rvert$	0	0.0114
Lemma 1(b) residual $\lvert C_v' - \partial_v J_v\rvert$	0	0.0098
$\lvert C_v\rvert$, honest vs. deceptive (Lemma 2)	equal	0.01007 vs. 0.00970
D_v deceptive (Theorem 3)	0.2002	0.2057
D_v honest	η	0.0068

The two regimes are separated by a factor of 30 at $\rho = 0.03$, while $|C_v|$ is statistically indistinguishable between them, confirming Lemma 2. The identity (12) holds during the transient as well as at steady state (residual 0.0041 vs. 0.0056). Adding the second probe displaces the DNE by 3×10^{-5}.

Sweeping $a \in \{0.025, 0.05, 0.1\}$, $k \in \{0.1, 0.3\}$ and scaling all frequencies by $1, 2, 4$ gives the honest-side floor

$$\eta \;\approx\; 0.19 \left(\frac{k}{a\,\omega_{\min}}\right)^2$$

with fit error below 5% and no dependence on the integration step, confirming (16)(ii) and the reversal of the naive rule "take a small": halving a from 0.05 to 0.025 raises η from 0.0068 to 0.0274, close to the threshold $\rho = 0.03$. Reducing the window to $T = 2$ raises the honest peak to $0.0825 > \rho$ even with noiseless measurements, in agreement with Remark 7.

Two further predictions. Proposition 5 and Corollary 7 are directly testable on the same setup and should be reported alongside the table above. *(i) Evasion.* Running the deceiver with a mirrored injection $\delta' \equiv \delta_d$ at ω_v' should collapse D_v to the honest floor $\eta \approx 0.0068$ while the DNE action x_v^* moves by $O(a^2)$ only; sweeping $\delta' \in [0, 2\delta_d]$ should trace the V-shape $D_v = |\delta' - \delta_d|\,|\partial_d J_v(u^*)|$ with slope $|\partial_d J_v(u^*)| = c\,x_v^* + e \approx 0.663$. *(ii) Windup.* The honest NE of this game is $u^\star = (0.7692, 0.7692)$ with $J_d(u^\star) = 0.2041$, so $\gamma = J_d(u^\star) - J_d^{\mathrm{ref}} = 0.0541$ and Corollary 7 predicts that after the frequency switch an unsaturated δ_d grows linearly at rate $\varepsilon\gamma \approx 4.33 \times 10^{-3}$ per unit time, while x_v converges to 0.7692 regardless. Terminating the run when $a|\delta_d|$ exceeds a stipulated actuation limit gives the time-to-visibility that Remark 9 appeals to.

References

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